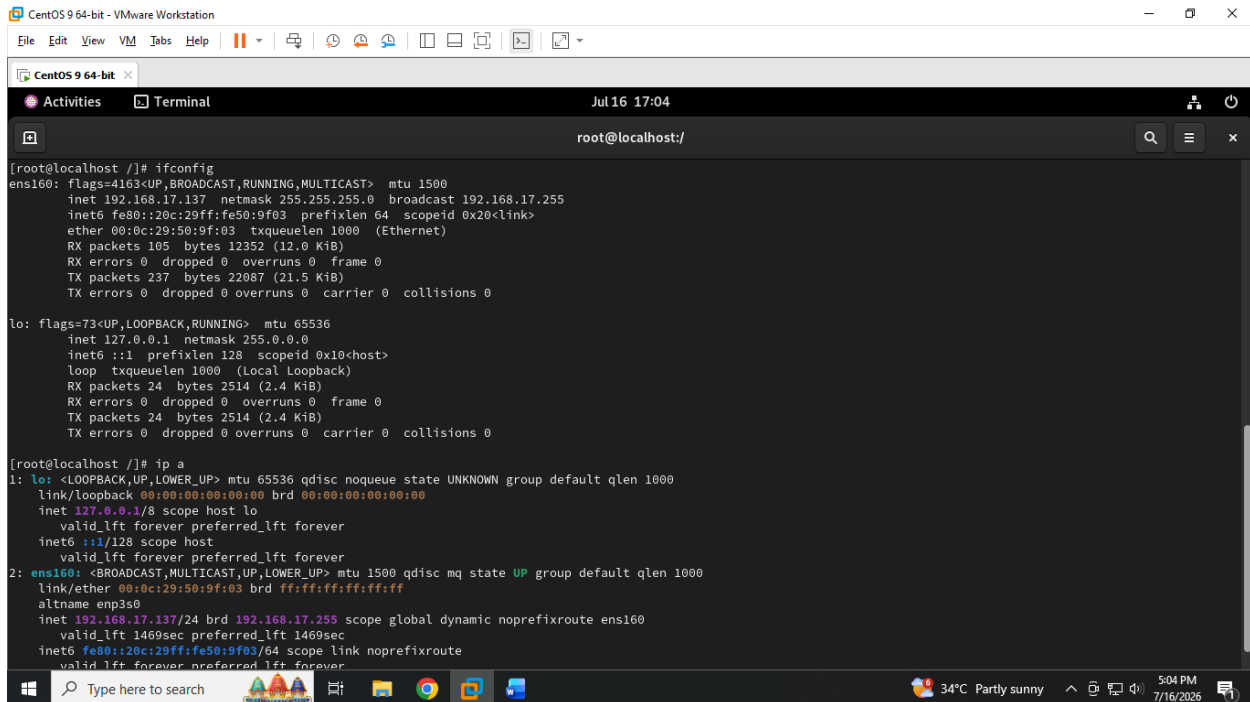


MODULE-10 : CONFIGURING NETWORK INTERFACES & UNDERSTANDING DNS, DHCP & FIREWALL MANAGEMENT

GUIDED BY : VIPUL MISTRI

Q.1 Use the ifconfig or ip command to display the current IP address and network interfaces on your Linux system, then take a screenshot of the output.

ANS :



```
CentOS 9 64-bit - VMware Workstation
File Edit View VM Tabs Help
CentOS 9 64-bit x
Activities Terminal Jul 16 17:04
root@localhost:/

[root@localhost ~]# ifconfig
ens160: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.17.137 netmask 255.255.255.0 broadcast 192.168.17.255
    inet6 fe80::20c:29ff:fe50:9f03 prefixlen 64 scopeid 0x20<link>
    ether 00:0c:29:50:9f:03 txqueuelen 1000 (Ethernet)
    RX packets 105 bytes 12352 (12.0 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 237 bytes 22087 (21.5 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 24 bytes 2514 (2.4 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 24 bytes 2514 (2.4 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

[root@localhost ~]# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: ens160: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen 1000
    link/ether 00:0c:29:50:9f:03 brd ff:ff:ff:ff:ff:ff
    altname enp3s0
    inet 192.168.17.137/24 brd 192.168.17.255 scope global dynamic noprefixroute ens160
        valid_lft 1469sec preferred_lft 1469sec
    inet6 fe80::20c:29ff:fe50:9f03/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
```

MODULE-10 : CONFIGURING NETWORK INTERFACES & UNDERSTANDING DNS, DHCP & FIREWALL MANAGEMENT

GUIDED BY : VIPUL MISTRI

Q.2 Configure a static IP address for your system using nmcli so that your device always gets the same IP (for example, 192.168.1.50/24 with gateway 192.168.1.1 and DNS 8.8.8.8). Hint: Use 'nmcli con mod' to set ipv4.addresses, ipv4.gateway, and ipv4.dns, then bring the connection down and up to apply changes.

ANS :

```
CentOS 9 64-bit - VMware Workstation
File Edit View VM Tabs Help
CentOS 9 64-bit x
Activities Terminal Jul 16 17:04
root@localhost:/

[root@localhost ~]# ifconfig
ens160: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.17.137 netmask 255.255.255.0 broadcast 192.168.17.255
    inet6 fe80::20c:29ff:fe50:9f03 prefixlen 64 scopeid 0x20<link>
    ether 00:0c:29:50:9f:03 txqueuelen 1000 (Ethernet)
    RX packets 105 bytes 12352 (12.0 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 237 bytes 22087 (21.5 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 24 bytes 2514 (2.4 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 24 bytes 2514 (2.4 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

[root@localhost ~]# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: ens160: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen 1000
    link/ether 00:0c:29:50:9f:03 brd ff:ff:ff:ff:ff:ff
    altname enp3s0
    inet 192.168.17.137/24 brd 192.168.17.255 scope global dynamic noprefixroute ens160
        valid_lft 1469sec preferred_lft 1469sec
    inet6 fe80::20c:29ff:fe50:9f03/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
```

```
CentOS 9 64-bit - VMware Workstation
File Edit View VM Tabs Help
CentOS 9 64-bit x
Activities Terminal Jul 16 17:10
root@localhost:/

[root@localhost ~]# nmcli connection modify ens160 ipv4.method manual ipv4.addresses 192.168.1.50/24
[root@localhost ~]# nmcli connection modify ens160 ipv4.gateway 192.168.1.1
[root@localhost ~]# nmcli connection modify ens160 ipv4.dns 8.8.8.8
[root@localhost ~]# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: ens160: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen 1000
    link/ether 00:0c:29:50:9f:03 brd ff:ff:ff:ff:ff:ff
    altname enp3s0
    inet 192.168.17.137/24 brd 192.168.17.255 scope global dynamic noprefixroute ens160
        valid_lft 1163sec preferred_lft 1163sec
    inet6 fe80::20c:29ff:fe50:9f03/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
[root@localhost ~]# nmcli connection down ens160
Connection 'ens160' successfully deactivated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/2)
[root@localhost ~]# nmcli connection up ens160
Connection successfully activated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/3)
[root@localhost ~]# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: ens160: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen 1000
    link/ether 00:0c:29:50:9f:03 brd ff:ff:ff:ff:ff:ff
    altname enp3s0
    inet 192.168.1.50/24 brd 192.168.1.255 scope global noprefixroute ens160
        valid_lft forever preferred_lft forever
```

MODULE-10 : CONFIGURING NETWORK INTERFACES & UNDERSTANDING DNS, DHCP & FIREWALL MANAGEMENT

GUIDED BY : VIPUL MISTRI

Q.3 Explain in your own words the difference between DNS and DHCP, and give a real-world example of how each is used in an app like Zomato or Flipkart.

ANS : DNS---- DNS (Domain Name Server) is a server which is used to translate website name into ip address eg: www.google.com-> 10.0.0.1 . without DNS when you want to visit any website you have to manually write the ip address of the website, because server and network using which you are accessing websites only understand ip addresses.

DHCP---- DHCP(Dynamic Host Configuration Protocol) DHCP is a protocol that helps to give ip addresses to the devices automatically. When you have too many devices it is not possible to give ip address to the device one by one manually, so here we can use DHCP.

Real world example :

Dns: when you open zomato's website in you pc or laptop or phone you search of Zomato.com or Zomato. Here you write in human readable form but machine did not understand this form, if there is no DNS then you have to manually enter the ip address of the Zomato eg : you go to browser and type 10.0.0.10.

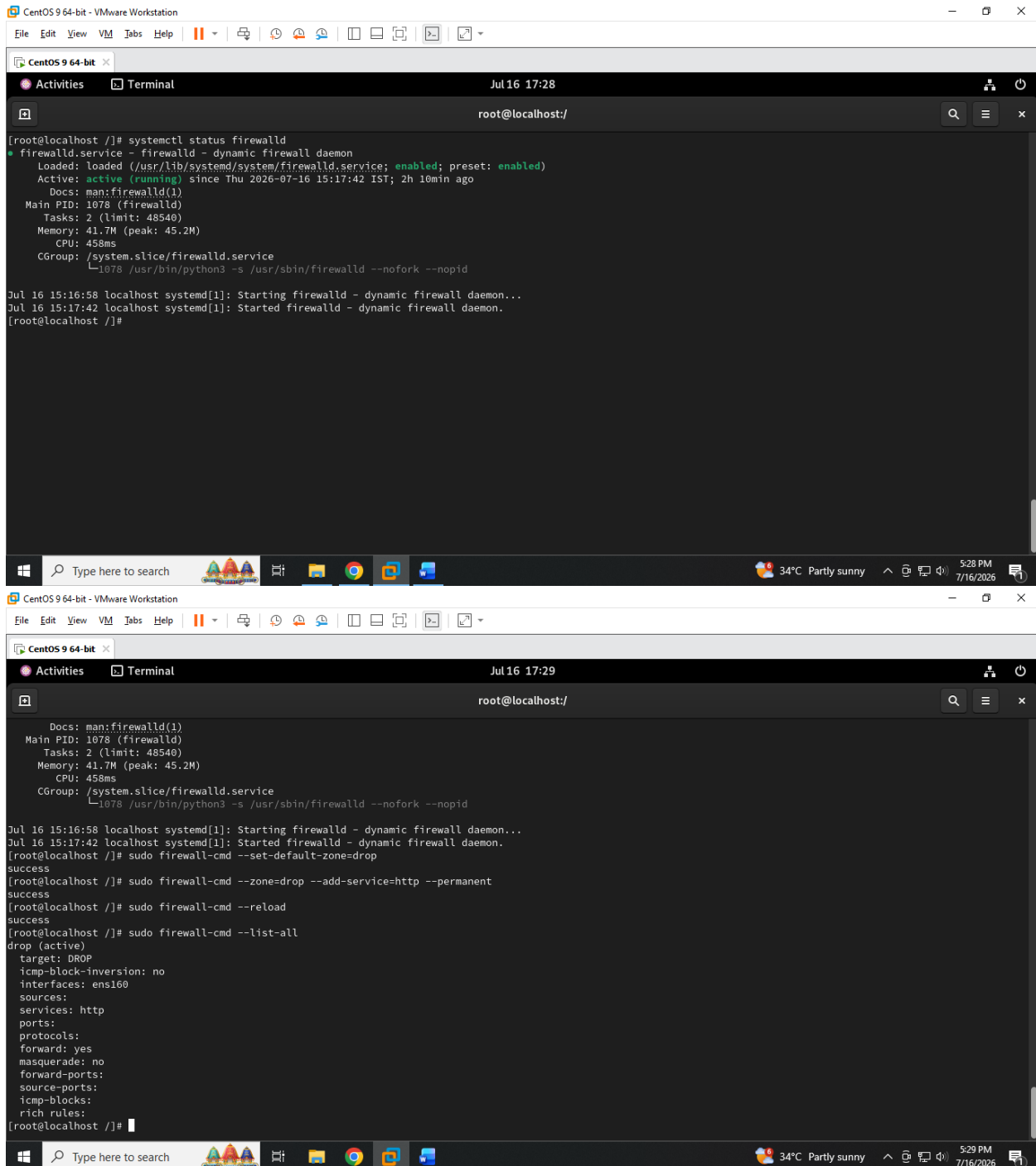
DHCP : when you open website before that if you want to use website you will need internet connection to access the website and when you connect to wifi or you internet the DHCP server automatically give you an ip address . in order to use internet you need an ip address either manual or automatic.

MODULE-10 : CONFIGURING NETWORK INTERFACES & UNDERSTANDING DNS, DHCP & FIREWALL MANAGEMENT

GUIDED BY : VIPUL MISTRI

Q.4 Enable the firewall on your system using ufw, then allow incoming connections only on port 80 (HTTP) and block all others. Show the status output after applying these rules.

ANS :



```
CentOS 9 64-bit - VMware Workstation
File Edit View VM Tabs Help
CentOS 9 64-bit x
Activities Terminal Jul 16 17:28
root@localhost:/

[root@localhost ~]# systemctl status firewalld
● firewalld.service - firewalld - dynamic firewall daemon
   Loaded: loaded (/usr/lib/systemd/system/firewalld.service; enabled; preset: enabled)
   Active: active (running) since Thu 2026-07-16 15:17:42 IST; 2h 10min ago
     Docs: man:firewalld(1)
   Main PID: 1078 (firewalld)
    Tasks: 2 (limit: 48540)
   Memory: 41.7M (peak: 45.2M)
      CPU: 458ms
   CGroup: /system.slice/firewalld.service
           └─1078 /usr/bin/python3 -s /usr/sbin/firewalld --nofork --nopid

Jul 16 15:16:58 localhost systemd[1]: Starting firewalld - dynamic firewall daemon...
Jul 16 15:17:42 localhost systemd[1]: Started firewalld - dynamic firewall daemon.
[root@localhost ~]#

CentOS 9 64-bit - VMware Workstation
File Edit View VM Tabs Help
CentOS 9 64-bit x
Activities Terminal Jul 16 17:29
root@localhost:/

Docs: man:firewalld(1)
Main PID: 1078 (firewalld)
Tasks: 2 (limit: 48540)
Memory: 41.7M (peak: 45.2M)
CPU: 458ms
CGroup: /system.slice/firewalld.service
        └─1078 /usr/bin/python3 -s /usr/sbin/firewalld --nofork --nopid

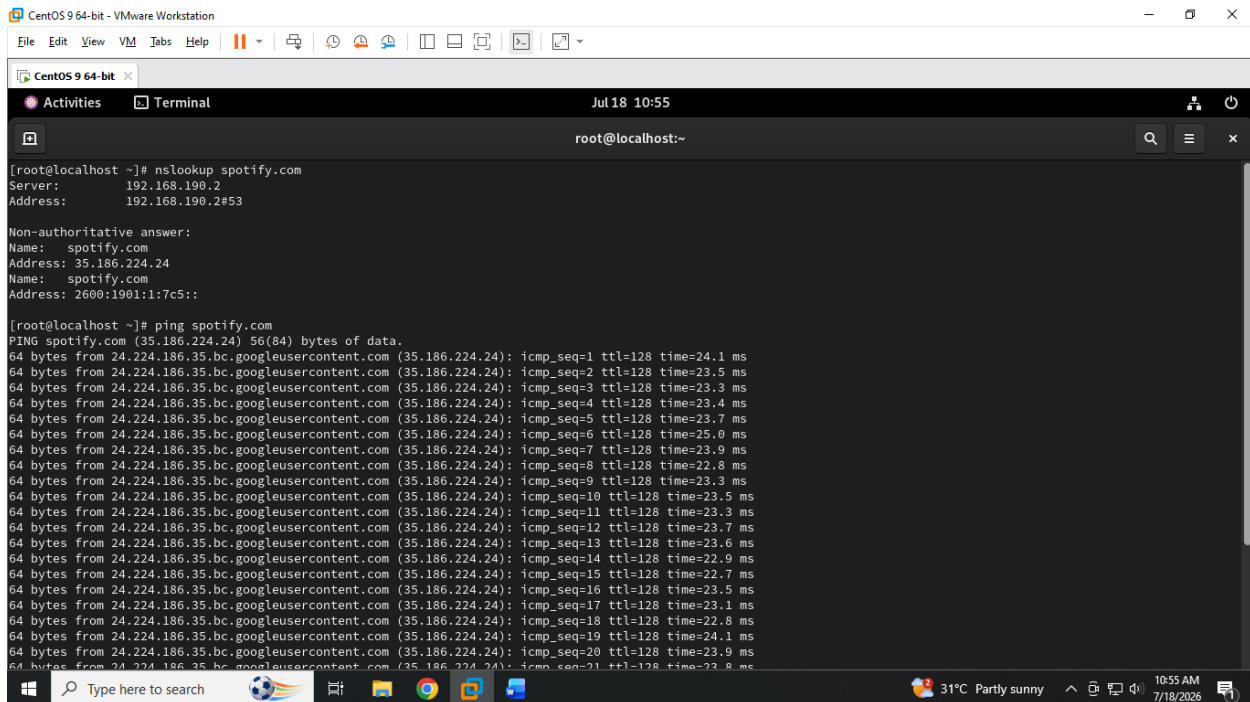
Jul 16 15:16:58 localhost systemd[1]: Starting firewalld - dynamic firewall daemon...
Jul 16 15:17:42 localhost systemd[1]: Started firewalld - dynamic firewall daemon.
[root@localhost ~]# sudo firewall-cmd --set-default-zone=drop
success
[root@localhost ~]# sudo firewall-cmd --zone=drop --add-service=http --permanent
success
[root@localhost ~]# sudo firewall-cmd --reload
success
[root@localhost ~]# sudo firewall-cmd --list-all
drop (active)
target: DROP
icmp-block-inversion: no
interfaces: ens160
sources:
services: http
ports:
protocols:
forward: yes
masquerade: no
forward-ports:
source-ports:
icmp-blocks:
rich rules:
[root@localhost ~]#
```

MODULE-10 : CONFIGURING NETWORK INTERFACES & UNDERSTANDING DNS, DHCP & FIREWALL MANAGEMENT

GUIDED BY : VIPUL MISTRI

Q.5 Use firewalld or iptables to temporarily block all outgoing traffic to a specific website (for example, www.spotify.com) and then verify that you cannot ping the site from your terminal. Hint: You may need to use the site's IP address instead of the domain name.

ANS :



```
CentOS 9 64-bit - VMware Workstation
File Edit View VM Tabs Help
CentOS 9 64-bit x
Activities Terminal Jul 18 10:55
root@localhost:~

[root@localhost ~]# nslookup spotify.com
Server:      192.168.190.2
Address:     192.168.190.2#53

Non-authoritative answer:
Name:   spotify.com
Address: 35.186.224.24
Name:   spotify.com
Address: 2600:1901:1:7c5::

[root@localhost ~]# ping spotify.com
PING spotify.com (35.186.224.24) 56(84) bytes of data:
64 bytes from 24.224.186.35.bc.googleusercontent.com (35.186.224.24): icmp_seq=1 ttl=128 time=24.1 ms
64 bytes from 24.224.186.35.bc.googleusercontent.com (35.186.224.24): icmp_seq=2 ttl=128 time=23.5 ms
64 bytes from 24.224.186.35.bc.googleusercontent.com (35.186.224.24): icmp_seq=3 ttl=128 time=23.3 ms
64 bytes from 24.224.186.35.bc.googleusercontent.com (35.186.224.24): icmp_seq=4 ttl=128 time=23.4 ms
64 bytes from 24.224.186.35.bc.googleusercontent.com (35.186.224.24): icmp_seq=5 ttl=128 time=23.7 ms
64 bytes from 24.224.186.35.bc.googleusercontent.com (35.186.224.24): icmp_seq=6 ttl=128 time=25.0 ms
64 bytes from 24.224.186.35.bc.googleusercontent.com (35.186.224.24): icmp_seq=7 ttl=128 time=23.9 ms
64 bytes from 24.224.186.35.bc.googleusercontent.com (35.186.224.24): icmp_seq=8 ttl=128 time=22.8 ms
64 bytes from 24.224.186.35.bc.googleusercontent.com (35.186.224.24): icmp_seq=9 ttl=128 time=23.3 ms
64 bytes from 24.224.186.35.bc.googleusercontent.com (35.186.224.24): icmp_seq=10 ttl=128 time=23.5 ms
64 bytes from 24.224.186.35.bc.googleusercontent.com (35.186.224.24): icmp_seq=11 ttl=128 time=23.3 ms
64 bytes from 24.224.186.35.bc.googleusercontent.com (35.186.224.24): icmp_seq=12 ttl=128 time=23.7 ms
64 bytes from 24.224.186.35.bc.googleusercontent.com (35.186.224.24): icmp_seq=13 ttl=128 time=23.6 ms
64 bytes from 24.224.186.35.bc.googleusercontent.com (35.186.224.24): icmp_seq=14 ttl=128 time=22.9 ms
64 bytes from 24.224.186.35.bc.googleusercontent.com (35.186.224.24): icmp_seq=15 ttl=128 time=22.7 ms
64 bytes from 24.224.186.35.bc.googleusercontent.com (35.186.224.24): icmp_seq=16 ttl=128 time=23.5 ms
64 bytes from 24.224.186.35.bc.googleusercontent.com (35.186.224.24): icmp_seq=17 ttl=128 time=23.1 ms
64 bytes from 24.224.186.35.bc.googleusercontent.com (35.186.224.24): icmp_seq=18 ttl=128 time=22.8 ms
64 bytes from 24.224.186.35.bc.googleusercontent.com (35.186.224.24): icmp_seq=19 ttl=128 time=24.1 ms
64 bytes from 24.224.186.35.bc.googleusercontent.com (35.186.224.24): icmp_seq=20 ttl=128 time=23.9 ms
64 bytes from 24.224.186.35.bc.googleusercontent.com (35.186.224.24): icmp_seq=21 ttl=128 time=23.8 ms
```

sudo firewall-cmd --permanent \

--add-rich-rule='rule family="ipv4" destination address="35.186.224.24" reject'

sudo firewall-cmd --reload